AHOPPA: Agentic Humanoid Operation through Perception, Planning, and Action

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Project page and code Raw data

Abstract: We study online humanoid operation with frontier multimodal models and report results obtained in the last few weeks following their release. AHOPPA connects model decisions to the fixed SONIC controller on a simulated Unitree G1. The model uses cameras, a static map and a revisable plan to inspect four cabinet drawers, push open drawers shut by hand and return home. It chooses routes and generates numerical head and wrist targets with locomotion commands. Physics continues during inference. A simulator evaluator checks completion. Across 64 scheduled episodes with four models at low and high reasoning effort, GPT-6 Astra completed 12 of 14 task attempts, taking 615 to 1492 s. These episodes show extended visual inspection, contact and return-home behavior through a fixed controller without a task skill library. Falls, timeouts and incorrect completion claims expose limits of this capability in simulation.

Keywords: Humanoid control, multimodal agents, online evaluation

1 Online humanoid operation

Extended tasks require a humanoid to decide where to look, how to move and when to stop. We test frontier models on a cabinet inspection task using visual feedback and numerical SONIC references [1].

Prior work grounds skills [2], generates code [3], uses feedback [4], builds value maps [5] or scene-graph plans [6], selects visual actions [7], uses keypoint constraints [8] or learns action tokens [9]. Embodied First Exam tests coding agents in continuous time [10]. Drawer closing is established [11]. Humanoids sequence skills [12, 13] or generate policy code [14].

2 Task, interface and evaluation

Scene. Newton physics [15] in Isaac Lab [16] runs at 200 Hz with native SONIC at 50 Hz. Four NVIDIA Sektion cabinets from Isaac Lab's cabinet task [17] stand in NVIDIA's Simple Room [18, 19]: A and B on the far wall behind a 2.1 m block of SimReady storage lockers [20], C and D behind home, so no drawer front is visible from the start. Each top drawer is a passive slide that closes only through contact, with 5 N dry friction and critical damping at its stop. The robot stands on Isaac Lab's flat ground collider beneath the room's visual floor. Eight conditions open none, one or two drawers (0.20 m) under an identical map and instruction.

Interface. The model emits four head/wrist pose keyframes with SONIC's walking and posture commands, and binary Dex3 hand presets. It chooses route, speed, facing, reach and timing; SONIC supplies gait, balance and tracking. The interface holds facing commands within 15° of the measured heading during turns, following the released gamepad. There is no drawer-closing, navigation or inspection skill. Each decision receives onboard RGB-D rendered from the measured torso pose, a fixed external RGB view, robot-only kinematics, depth queries, the facility map, its plan and recent



Figure 1: Exact model inputs from one successful GPT-6 Astra (low) episode with A and D open: the static map and images at the marked request times.

Model	Success	Physical	Closed	Seen	False	Control	Est. $
GPT-6 Astra (low)	6/7	6	9	5	0	1/1	9.30–16.12
GPT-6 Astra (high)	6/7	6	8	6	0	0/1	7.65–12.30
GPT-6.1 Sol (low)	4/7	4	4	5	1	0/1	1.71–2.99*
GPT-6.1 Sol (high)	3/7	5	9	6	0	0/1	1.72–2.86*
Claude Opus 5.5 (low)	2/7	3	5	4	2	1/1	4.76*
Claude Opus 5.5 (high)	3/7	3	6	3	1	1/1	7.22*
Gemini 4 Argon (low)	0/7	0	0	1	0	0/1	1.29*
Gemini 4 Argon (high)	0/7	0	2	1	0	0/1	1.59*

Table 1: Successes over seven scheduled task attempts; the all-closed control is separate. Other counts use completed episodes. *Physical*: closed, home and upright; *Seen*: all drawer fronts viewed; *False*: declarations without physical completion. *Est. $*: mean API cost per attempt at published rates [21, 22, 23]. Ranges span cache-write prices; * excludes unrecorded usage; GPU costs are excluded. Six API-error starts remain in success denominators.

action feedback. Simulator positions, drawer joints, contacts and evaluator state remain private to the simulator.

Scoring. A drawer is closed when within 2 cm of closed and still for 1 s. Filtered normal-force sensors separate hand pushes from torso or other contact. Success requires every drawer closed, a supported standing hold within 25 cm of home, and the model's completion declaration. Physical completion without a declaration, disturbed closed drawers, hand attribution, falls, timeouts and infrastructure errors are reported separately, as are controls with no open drawer.

Protocol. After floor and facing pilots, we froze scene, cameras, map, prompts, evaluator, an 1800 s budget and source, then ran one seeded episode per condition for each of GPT-6.1 Sol, GPT-6 Astra, Claude Opus 5.5 and Gemini 4 Argon at low and high reasoning effort, four episodes at a time with pinned host placement; all starts are retained.

3 Results

All 64 scheduled starts are retained (Table 1, Fig. 1). Every one of the 43 closures was by hand and no initially closed drawer was disturbed. Astra completed 6/7 episodes at both efforts (615 to 1492 s), usually checking every drawer, including the two behind the lockers. Sol completed 4/7 and 3/7, Opus 2/7 and 3/7, and Gemini none; high effort lengthened decisions (median 21 to 130 s versus 9 to 28 s) without consistent gains, and eight episodes ran out of time. Fifteen of 51 completed episodes starting with open drawers ended in falls, one at start-up before any model command; four declarations were made without the physical criterion.

Timing. Median peak physics lag was 192 ms; 17 of 58 completed episodes, including 7 successes, met the 100 ms limit. Falls rarely followed lag spikes. The results include episodes that exceeded this timing limit.

Limits. We study one simulated layout, one drawer interaction and one episode per model, effort and condition. Friction, damping and ground support are modeling choices. Reliability, capability gains and transfer require repeated trials, older-model baselines and hardware experiments.

References

[1] Z. Luo, Y. Yuan, T. Wang, C. Li, F. Castañeda, S. Chen, Z.-A. Cao, J. Li, D. Minor, Q. Ben, J. Park, D. Sami, Z. Wang, X. Da, R. Ding, C. Hogg, L. Song, E. Lim, E. Jeong, T. He, H. Xue, W. Xiao, S. Yuen, J. Kautz, Y. Chang, U. Iqbal, L. Fan, and Y. Zhu. SONIC: Supersizing motion tracking for natural humanoid whole-body control. *Science Robotics*, 11(117):eaed4592, 2026. doi:10.1126/scirobotics.aed4592. URL https://nvlabs.github.io/GEAR-SONIC/.

[2] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, C. Fu, K. Gopalakrishnan, K. Hausman, A. Herzog, D. Ho, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, E. Jang, R. J. Ruano, K. Jeffrey, S. Jesmonth, N. J. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, K.-H. Lee, S. Levine, Y. Lu, L. Luu, C. Parada, P. Pastor, J. Quiambao, K. Rao, J. Rettinghouse, D. Reyes, P. Sermanet, N. Sievers, C. Tan, A. Toshev, V. Vanhoucke, F. Xia, T. Xiao, P. Xu, S. Xu, M. Yan, and A. Zeng. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances, 2022. URL https://arxiv.org/abs/2204.01691.

[3] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng. Code as Policies: Language Model Programs for Embodied Control, 2023. URL https://arxiv.org/abs/2209.07753.

[4] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar, P. Sermanet, N. Brown, T. Jackson, L. Luu, S. Levine, K. Hausman, and B. Ichter. Inner Monologue: Embodied Reasoning through Planning with Language Models, 2022. URL https://arxiv.org/abs/2207.05608.

[5] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models, 2023. URL https://arxiv.org/abs/2307.05973.

[6] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suenderhauf. SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 23–72. PMLR, 2023. URL https://proceedings.mlr.press/v229/rana23a.html.

[7] S. Nasiriany, F. Xia, W. Yu, T. Xiao, J. Liang, I. Dasgupta, A. Xie, D. Driess, A. Wahid, Z. Xu, Q. Vuong, T. Zhang, T.-W. E. Lee, K.-H. Lee, P. Xu, S. Kirmani, Y. Zhu, A. Zeng, K. Hausman, N. Heess, C. Finn, S. Levine, and B. Ichter. PIVOT: Iterative visual prompting elicits actionable knowledge for VLMs. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 37321–37341. PMLR, 2024. URL https://proceedings.mlr.press/v235/nasiriany24a.html.

[8] W. Huang, C. Wang, Y. Li, R. Zhang, and L. Fei-Fei. Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 4573–4602. PMLR, 2025. URL https://proceedings.mlr.press/v270/huang25g.html.

[9] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, Q. Vuong, V. Vanhoucke, H. Tran, R. Soricut, A. Singh, J. Singh, P. Sermanet, P. R. Sanketi, G. Salazar, M. S. Ryoo, K. Reymann, K. Rao, K. Pertsch, I. Mordatch, H. Michalewski, Y. Lu, S. Levine, L. Lee, T.-W. E. Lee, I. Leal, Y. Kuang, D. Kalashnikov, R. Julian, N. J. Joshi, A. Irpan, B. Ichter, J. Hsu, A. Herzog, K. Hausman, K. Gopalakrishnan, C. Fu, P. Florence, C. Finn, K. A. Dubey, D. Driess, T. Ding, K. M. Choromanski, X. Chen, Y. Chebotar, J. Carbajal, N. Brown, A. Brohan, M. G. Arenas, and K. Han. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR, 2023. URL https://proceedings.mlr.press/v229/zitkovich23a.html.

[10] K. Wang, C. Chen, Q. Wang, P. Zhang, C. Gui, F. Wang, C. Chen, W. Xiao, X. Huang, T. Zhang, L. Li, Y. Sun, M. Deng, J. Hou, E. P. Xing, H. Yao, Z. Fan, Y. Liu, J. Wu, Y. Zhu, G. Shi, R. Zhang, and M. Li. Embodied first exam, 2026. URL https://github.com/mll-lab-nu/embodied-first-exam.

[11] S. James, Z. Ma, D. Rovick Arrojo, and A. J. Davison. RLBench: The robot learning benchmark & learning environment. *IEEE Robotics and Automation Letters*, 2020. URL https://github.com/stepjam/RLBench.

[12] G. Huh, C. Gu, T. E. Truong, C. K. Liu, and G. Tevet. HomeBody: A humanoid that explores, remembers, and acts on its own, 2026. URL https://tml.stanford.edu/homebody/.

[13] J. Sun, Q. Zhang, Y. Duan, X. Jiang, C. Cheng, and R. Xu. Prompt, Plan, Perform: LLM-based Humanoid Control via Quantized Imitation Learning, 2024. URL https://arxiv.org/abs/2309.11359.

[14] S. Choi, S. Ye, N. Bui, A. Shrivastava, K. Ryu, D. Tirumala, M. Wulfmeier, and N. Mehr. HuGo: LLMs as whole-body policy code designers for humanoid loco-manipulation, 2026. URL https://arxiv.org/abs/2609.30594.

[15] The Newton Contributors. Newton: GPU-accelerated physics simulation for robotics and simulation research, 2025. URL https://github.com/newton-physics/newton.

[16] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrügg, N. Rudin, L. Wawrzyniak, M. Rakhsha, A. Denzler, E. Heiden, A. Borovicka, O. Ahmed, I. Akinola, A. Anwar, M. T. Carlson, J. Y. Feng, A. Garg, R. Gasoto, L. Gulich, Y. Guo, M. Gussert, A. Hansen, M. Kulkarni, C. Li, W. Liu, V. Makoviychuk, G. Malczyk, H. Mazhar, M. Moghani, A. Murali, M. Noseworthy, A. Poddubny, N. Ratliff, W. Rehberg, C. Schwarke, R. Singh, J. L. Smith, B. Tang, R. Thaker, M. Trepte, K. V. Wyk, F. Yu, A. Millane, V. Ramasamy, R. Steiner, S. Subramanian, C. Volk, C. Chen, N. Jawale, A. V. Kuruttukulam, M. A. Lin, A. Mandlekar, K. Patzwaldt, J. Welsh, H. Zhao, F. Anes, J.-F. Lafleche, N. Moënne-Loccoz, S. Park, R. Stepinski, D. V. Gelder, C. Amevor, J. Carius, J. Chang, A. H. Chen, P. de Heras Ciechomski, G. Daviet, M. Mohajerani, J. von Muralt, V. Reutskyy, M. Sauter, S. Schirm, E. L. Shi, P. Terdiman, K. Vilella, T. Widmer, G. Yeoman, T. Chen, S. Grizan, C. Li, L. Li, C. Smith, R. Wiltz, K. Alexis, Y. Chang, D. Chu, L. J. Fan, F. Farshidian, A. Handa, S. Huang, M. Hutter, Y. Narang, S. Pouya, S. Sheng, Y. Zhu, M. Macklin, A. Moravanszky, P. Reist, Y. Guo, D. Hoeller, and G. State. Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning, 2025. URL https://arxiv.org/abs/2511.04831.

[17] Isaac Lab Project Developers. *Available Environments: Isaac-Open-Drawer-Franka-v0*, 2026. URL https://isaac-sim.github.io/IsaacLab/main/source/overview/environments.html.

[18] NVIDIA. *Environment Assets*. NVIDIA, 2026. URL https://docs.isaacsim.omniverse.nvidia.com/latest/assets/usd_assets_environments.html.

[19] NVIDIA. *NVIDIA Isaac Sim Additional Software and Materials License*. NVIDIA, 2025. URL https://www.nvidia.com/en-us/agreements/enterprise-software/isaac-sim-additional-software-and-materials-license/.

[20] NVIDIA. *SimReady Assets*. NVIDIA, 2026. URL https://huggingface.co/datasets/nvidia/simready-assets.

[21] OpenAI. API Pricing and Prompt Caching, 2026. URL https://developers.openai.com/api/docs/pricing.

[22] Anthropic. Claude Opus 5.5: Pricing, 2026. URL https://platform.claude.com/docs/en/models/opus-5-5/overview.

[23] Google. Gemini API Pricing, 2026. URL https://ai.google.dev/gemini-api/docs/pricing.